

Experiment 9

Aim: To understand Docker Architecture and Container Life Cycle, install Docker and execute docker commands to manage images and interact with containers.

Theory:

Docker is a popular platform that enables developers to build, package, and deploy applications as lightweight, portable, and self-sufficient containers. These containers encapsulate all the necessary dependencies and libraries required for an application to run, ensuring consistency across different environments. Here is a theoretical overview of

Docker:

Containerization:

Docker utilizes containerization technology to create isolated environments for applications. Containers are lightweight, standalone, and executable packages that include everything needed to run an application, such as code, runtime, system tools, libraries, and settings. This isolation ensures that applications run consistently across different environments, from development to production.

Docker Engine:

At the core of Docker is the Docker Engine, which is responsible for building, running, and managing containers. It consists of the Docker daemon, which manages containers, images, networks, and volumes, and the Docker client, which allows users to interact with the daemon through the Docker API.

Docker Images:

Docker images are read-only templates used to create containers. They contain the application code, runtime, libraries, dependencies, and other files needed to run the application. Images are built using Dockerfiles, which are text files that define the steps needed to create the image.

Docker Containers:

Containers are instances of Docker images that are running as isolated processes on a host machine. They are lightweight, portable, and can be easily started, stopped, moved, and deleted. Containers provide a consistent environment for applications to run, regardless of the underlying infrastructure.

Benefits of Docker:

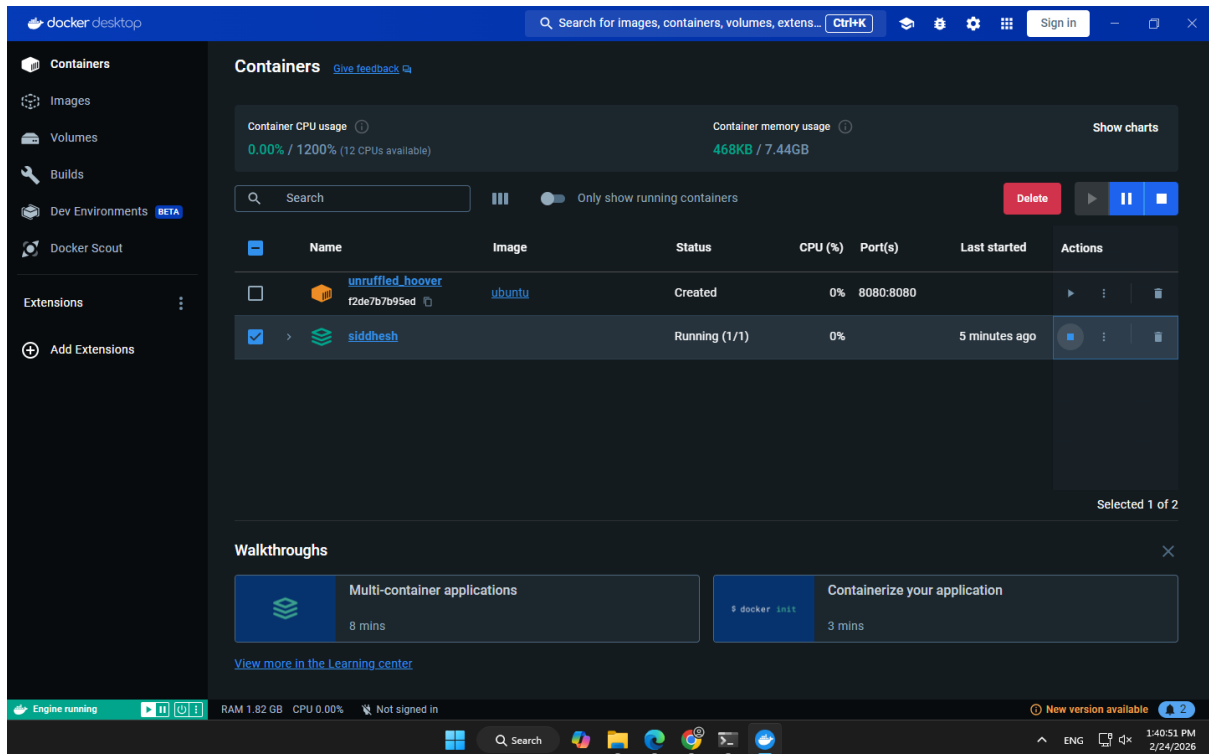
Portability: Docker containers can run on any platform that supports Docker, making it easy to deploy applications across different environments.

Efficiency: Containers share the host OS kernel, reducing overhead and improving resource utilization.

Isolation: Containers provide a level of isolation that helps prevent conflicts between applications and dependencies.

Scalability: Docker enables easy scaling of applications by quickly spinning up additional containers.

Consistency: Docker ensures that applications run the same way in development, testing, and production environments.



```
C:\Users\Lenovo>docker --help

Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
  run      Create and run a new container from an image
  exec     Execute a command in a running container
  ps       List containers
  build    Build an image from a Dockerfile
  pull     Download an image from a registry
  push     Upload an image to a registry
  images   List images
  login    Log in to a registry
  logout   Log out from a registry
  search   Search Docker Hub for images
  version  Show the Docker version information
  info     Display system-wide information

Management Commands:
  builder   Manage builds
  buildx*   Docker Buildx (Docker Inc., v0.12.1-desktop.4)
  compose*  Docker Compose (Docker Inc., v2.24.6-desktop.1)
  container Manage containers
  context   Manage contexts
  debug*    Get a shell into any image or container. (Docker Inc., 0.0.24)
  dev*      Docker Dev Environments (Docker Inc., v0.1.0)
  extension* Manages Docker extensions (Docker Inc., v0.2.22)
  feedback* Provide feedback, right in your terminal! (Docker Inc., v1.0.4)
  image     Manage images
  init*     Creates Docker-related starter files for your project (Docker Inc., v1.0.1)
  manifest  Manage Docker image manifests and manifest lists
  network   Manage networks
  plugin    Manage plugins
  sbom*     View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc., 0.6.0)
  scout*    Docker Scout (Docker Inc., v1.5.0)
  system    Manage Docker
  trust     Manage trust on Docker images
  volume    Manage volumes

C:\Users\Lenovo>docker --version
Docker version 25.0.3, build 4deb4f41

C:\Users\Lenovo>

C:\Users\Lenovo>docker search ubuntu

NAME                DESCRIPTION                                STARS    OFFICIAL
ubuntu              Ubuntu is a Debian-based Linux operating sys... 17789    [OK]
ubuntu/squid        Squid is a caching proxy for the Web. Long-t... 124
ubuntu/nginx        Nginx, a high-performance reverse proxy & we... 139
ubuntu/cortex       Cortex provides storage for Prometheus. Long... 4
ubuntu/bind9        BIND 9 is a very flexible, full-featured DNS... 117
ubuntu/kafka        Apache Kafka, a distributed event streaming ... 60
ubuntu/zookeeper    ZooKeeper maintains configuration informatio... 14
ubuntu/apache2      Apache, a secure & extensible open-source HT... 102
ubuntu/prometheus   Prometheus is a systems and service monitori... 78
ubuntu/mysql        MySQL open source fast, stable, multi-thread... 72
ubuntu/aspnet       Chiselled Ubuntu runtime image for ASP.NET a... 26
ubuntu/jre          Chiseled Java runtime based on Ubuntu. Long-... 24
ubuntu/postgres     PostgreSQL is an open source object-relatio... 42
ubuntu/python       A chiselled Ubuntu rock with the Python runt... 31
ubuntu/redis        Redis, an open source key-value store. Long-... 24
ubuntu/dotnet-runt... Chiselled Ubuntu runtime image for .NET apps... 24
ubuntu/dotnet-deps  Chiselled Ubuntu for self-contained .NET & A... 16
ubuntu/grafana       Grafana, a feature rich metrics dashboard & ... 12
ubuntu/memcached    Memcached, in-memory keyvalue store for smal... 6
ubuntu/alertmanag... Alertmanager handles client alerts from Prom... 10
ubuntu/cassandra    Cassandra, an open source NoSQL distributed ... 3
ubuntu/mlflow       MLFlow: for managing the machine learning li... 6
ubuntu/loki         Grafana Loki, a log aggregation system like ... 2
ubuntu/mimir       Ubuntu ROCK for Mimir, a horizontally scalab... 0
ubuntu/openteleme... This is a rock for the OpenTelemetry Collect... 0

C:\Users\Lenovo>docker search --filter=stars=1000 ubuntu

NAME                DESCRIPTION                                STARS    OFFICIAL
ubuntu              Ubuntu is a Debian-based Linux operating sys... 17789    [OK]
```

```
C:\Users\Lenovo>docker search mysql
NAME                DESCRIPTION                STARS    OFFICIAL
mysql                MySQL is a widely used, open-source relation... 16070    [OK]
bitnami/mysql        Bitnami Secure Image for mysql 147
circleci/mysql        MySQL is a widely used, open-source relation... 32
bitnamicharts/mysql   Bitnami Helm chart for MySQL 0
cimg/mysql            3
ubuntu/mysql          MySQL open source fast, stable, multi-thread... 72
google/mysql          MySQL server for Google Compute Engine 26
linuxserver/mysql     A Mysql container, brought to you by LinuxSe... 48
elestio/mysql         Mysql, verified and packaged by Elestio 3
alpine/mysql          mysql client 7
docksal/mysql         MySQL service images for Docksal - https://d... 0
eclipse/mysql         Mysql 5.7, curl, rsync 3
ilios/mysql           Mysql configured for running Ilios 1
datajoint/mysql       MySQL image pre-configured to work smoothly ... 2
ddev/mysql            ARM64 base images for ddev-dbserver-mysql-8... 2
mirantis/mysql        0
mysql/mysql-server    Optimized MySQL Server Docker images. Create... 1033
corpusops/mysql       https://github.com/corpusops/docker-images/ 0
cleanstart/mysql      Secure by Design, Built for Speed, Hardened ... 0
vulhub/mysql         1
cbioportal/mysql      This repository hosts MySQL database images ... 1
vites/mysql          Lightweight image to run MySQL with Vitess 1
mysql/mysql-router     MySQL Router provides transparent routing be... 28
mysql/mysql-cluster   Experimental MySQL Cluster Docker images. Cr... 100
nasqueron/mysql        1

C:\Users\Lenovo>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
01d7766a2e4a: Pull complete
Digest: sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4f9
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

What's Next?
1. Sign in to your Docker account → docker login
2. View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu

C:\Users\Lenovo>
C:\Users\Lenovo>docker images
REPOSITORY          TAG                IMAGE ID           CREATED            SIZE
ubuntu              latest            bbdabce66f1b      13 days ago       78.1MB
ubuntu              <none>            802541663949      6 months ago      78.1MB
ubuntu              <none>            ca2b0f26964c      24 months ago     77.9MB
redis               <none>            170a1e90f843      2 years ago       138MB
docker/dev-environments-default stable-1           7eb85f44c229      3 years ago       610MB
crazyman/linguist   7.20.0            dc5f9ae3be6a      3 years ago       72.2MB
docker/desktop-git-helper 5a4fcal26aadcd3f6cc3a011aa991de982ae7000 efe2d67c403b      4 years ago       44.2MB

C:\Users\Lenovo>docker run -it ubuntu
root@715e3a4a7ed8:/# exit
exit

C:\Users\Lenovo>docker run -it ubuntu
root@2ddd2c7718ec:/# admin
bash: admin: command not found
root@2ddd2c7718ec:/# ^C
root@2ddd2c7718ec:/# exit
exit

C:\Users\Lenovo>docker run -d httpd
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
0c8d55a45c0d: Pull complete
fe1db8dd446f: Pull complete
4f4fb700ef54: Pull complete
30973b009bab: Pull complete
85b0d820aa56: Pull complete
996228ca33ab: Pull complete
Digest: sha256:b89c19a390514d6767e8c62f29375d0577190be448f63b24f5f11d6b03f7bf18
Status: Downloaded newer image for httpd:latest
9f106d7c64c6c3703dc235fe26ca50a03a9876c261d5b29b20a30b006922393d

C:\Users\Lenovo>docker run -d mysql
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
74a8e4bbd9fe: Pull complete
357926362d31: Pull complete
f3702ac323ca: Pull complete
658630c937c6: Pull complete
ab1b8ea72826: Pull complete
61941d436e88: Pull complete
c89a19cdad3b: Pull complete
00af5f73db53: Pull complete
8314ff001dec: Pull complete
4a629f1008ff: Pull complete
Digest: sha256:e5dc14f6e01c3e577e669337d2855c6d1561b30d8ef2c592e63e4e8a9a52650a
Status: Downloaded newer image for mysql:latest
af5bbbf473e08d0b9a7264168926d9712c94a6d94497687fed89179d31e770ccd

C:\Users\Lenovo>docker run -d ubuntu
17ae4f0c3240c5b5ba81df3d23be8a1eef290d48684f89b47335567685514b7a

C:\Users\Lenovo>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
9f10642c64c6   httpd     "httpd-foreground"      About a minute ago Up About a minute 80/tcp       wonderful_babbage
a388ed3e2ae7   docker/dev-environments-default:stable-1 "sleep infinity"        23 months ago Up 12 minutes 80/tcp       siddhesh-app-1
```